

EDEM Physics Tutorials

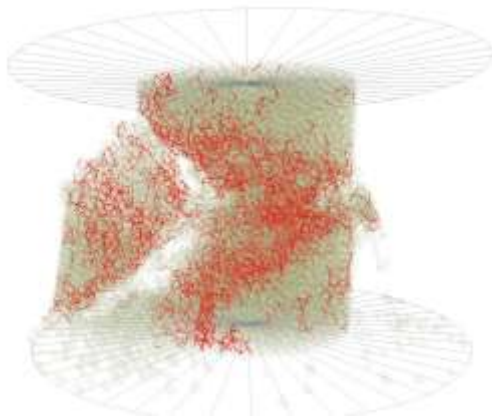
Linear Elastic Bond Model (LEBM)

Introduction

This exercise explores additional options and models of EDEM, focusing on the use of the Linear Elastic Bonding Model combined with Volume Packing.

The focus of this tutorial is on:

- Using the LEBM contact model
- Setting up and defining the particle contact radius
- Using Volume Packing
- Displaying and analyzing the bonds between particles



There are two types of Bond model available in EDEM:

The Bonding V2 Model

The Bonding V2 model is an 'Additional' model that can be added to any Base model. At a set point in time, or on particle creation, bonds are created between particle elements. The bonds apply normal and tangential forces to the particle in addition to rotational torque. Forces and torques are applied up to a user defined limit after which the bond will break. Details of the Bonding V2 model can be seen in the EDEM Help > EDEM Creator > Creator Tree-Physics > **Additional Models** > The Bonding V2 Model.

The Bonding V2 Model is typically used for modelling rock and flexible materials.

The Linear Elastic Bonding Model (LEBM)

The LEBM model is an alternative to the Bonding V2 model. The LEBM model is a combined model which contains the base physics in addition to the bonding forces.

At the start of the simulation bonds are created between the particle and apply normal and tangential (shear) forces to the bonded particles, up to the pre-defined bond strength limits. Details of the Linear Elastic Bonding Model can be found in the EDEM Help > EDEM Creator > Creator Tree-Physics > **Plug-in Models** > The Linear Elastic Bonding Model.

The LEBM model is typically used for modeling rock and soil type materials.

EDEM Creator: Setting up the model

1. Start EDEM
2. Go to File > Save As...
3. Select a location on the local machine (e.g., **C:\EDEM_Tutorials**)
4. Enter a file name (e.g., **LEBM_setup.dem**) and click Save

Step 1: Set the Global Model Parameters

Choose the units

The first step in creating the model is to set the units used throughout EDEM.

1. Go to the Tools > Options... menu and select the Units tab.
2. Change the following measurement units:
 - Velocity to **m/s**
 - Length to **mm**
3. Click OK.

Enter the model title and description

1. Click on Project in the Creator Tree.
2. In the Detailed View, enter the title **LEBM Unconfined Compression** in the Title box in the simulation section.

The model title and description will appear in the Data Browser window.

Step 2: Define the Bulk Material

Add new Bulk Material

1. Right click Bulk Material in the Creator Tree and select **Add Bulk Material (BulkMaterial 1)**
2. Set the Poisson's ratio, Shear Modulus and Density:

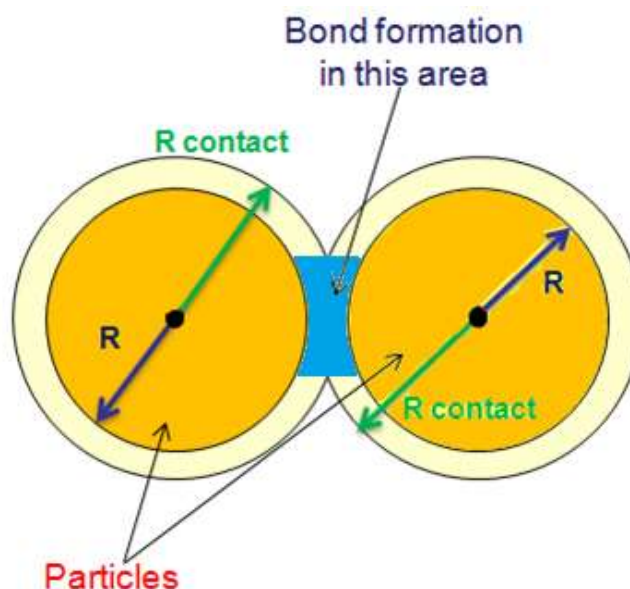
Poisson's Ratio (ν):	0.1
Solids Density (ρ):	3000 kg/m ³
☒ Shear Modulus (G):	1.5e+06 Pa
☐ Young's Modulus (E):	3.3e+06 Pa

3. Select **BulkMaterial 1** > Click the  button in the Interactions section
4. Select the material and leave the settings as default

Create a new particle type

In this exercise, a bonded particles model is used. When using bonded models, it is best to have a Contact Radius between the particles which is greater than the Physical Radius. The Contact Radius is important as it determines the area in which a bond can exist. When the particles no longer overlap the bond information is lost.

A bond will be formed in the first Time Step of the simulation only. After these initial bonds have been created, no new bonds will be formed. Due to the formation of bonds during the first Time Step, either the Volume Packing tool can be used which creates particles prior to the first time-step starting (as detailed in this tutorial) or export a previous simulation with particles already in place and the particles are created using a Contact Radius (where possible) to allow for bonds to be created between particles even when physically not in contact.



1. Right click on **BulkMaterial 1** > Add Shape from Library > Single Sphere
2. Rename the particle to **Bonded particle**

3. Set the Radius of the sphere to **1.5 mm**
4. Check the **Edit Contact Radius** box, situated below the Viewer Window. A new column will appear in the table below
5. Set the value of the Contact Radius to **1.7 mm**

Bonded particle Spheres					
☒ Edit Contact Radius					
Name	Position X (mm)	Position Y (mm)	Position Z (mm)	Physical Radius (mm)	Contact Radius (mm)
0 sphere 0	0	0	0	1.5	1.7

6. Set the size distribution:

Size Distribution

random

Random Size Distribution Parameters

Minimum: 0.8

Maximum: 1.2

Scale By

☒ Radius

☐ Volume

7. Go to Bonded particle > Properties and select Calculate Properties based on Spheres.

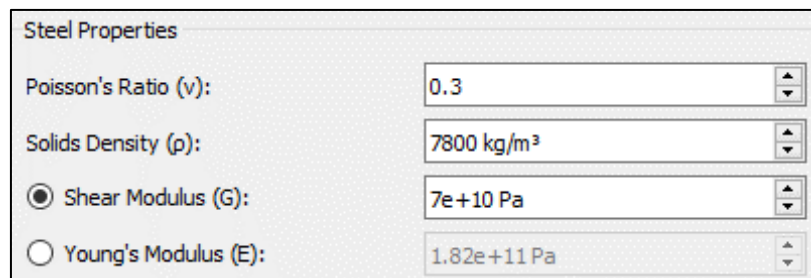
Step 3: Define the Equipment Material

Add new equipment material

1. Right click Equipment Material in the Creator Tree and select Add Equipment Material.

Define the equipment material properties

1. Set the Equipment Material name as **Steel** and define the Poisson's ratio, Shear Modulus and Density as shown below:



Steel Properties	
Poisson's Ratio (ν):	0.3
Solids Density (ρ):	7800 kg/m ³
☒ Shear Modulus (G):	7e+10 Pa
☐ Young's Modulus (E):	1.82e+11 Pa

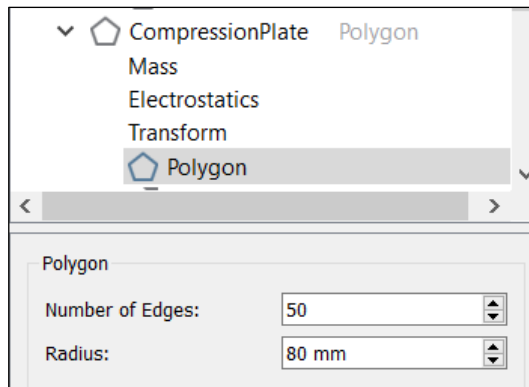
Define the material interaction properties

1. Click the  button in the Interactions section
2. Select **BulkMaterial 1** and then click OK
3. Leave the settings as default

Step 4: Define the Geometry

Create a Loading Plate

1. Right click on Geometries
2. Hover over Add Geometry and select **Polygon**
3. Rename this geometry to **CompressionPlate**
4. Set the Transform Z position to be 100 mm
5. Expand Polygon in the Creator Tree and set the Number of Edges and Radius as shown:



6. Right Click on CompressionPlate > Add Motion > Add Linear Translation Kinematic



Create the Base Plate

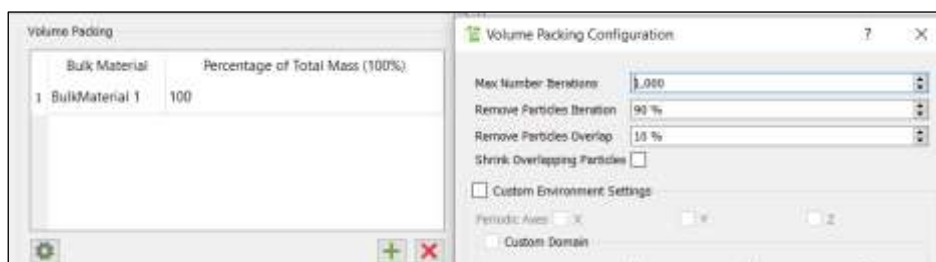
1. Right click on Geometries to create a second Polygon
2. Rename to **GroundPlate**
3. Set the number of edges to be 50 and radius to 80 mm
4. Leave the transform position to 0,0,0

Create the particle factory

1. Right click on Geometries to create a **Cylinder** and rename to **Particle factory**
2. Set the Type to be **Virtual**
3. Set the transform Z position to be 50 mm
4. Set the Radius start, end and Length as shown:



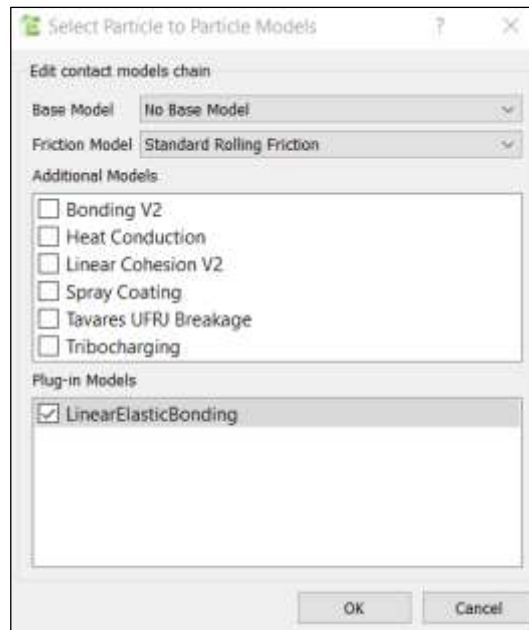
5. Right click on Particle factory to **Add Volume Packing**
6. Use  to add **BulkMaterial 1**
7. Set the **Imposed Solid Fraction** to be 60%
8. Set the **Start Time** to be 0s
9. Select the configuration option to de-select “shrink overlapping particles” as shown



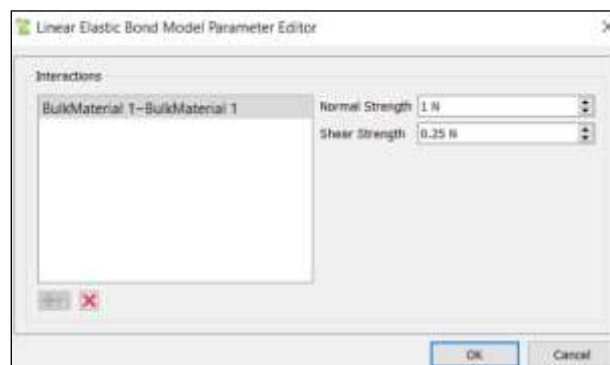
Step 5: Define the Physics

Set the contact models

1. Select the Physics section in the Creator Tree
2. Select **Particle to Particle** from the Interaction pull-down menu and click on **Edit Contact Chain**
3. Select **No Base Model** from the Base model list (this is to be replaced with the LinearElasticBonding model which includes the base component)
4. Choose **LinearElasticBonding** and select OK



5. Highlight the LinearElasticBonding model and click the configure button  to define the bonds
6. Click the  button and set the parameters as follows:



EDEM Simulator: Running a Simulation

Once the model has been set up in the Creator, we move to the Simulator to run it.

1. Click on the Simulator button  on the toolbar

Step 1: Set the Time Options

Set the time step

1. If checked, **uncheck** the Auto Time Step checkbox
2. Set the Fixed Time Step lower box to **1e-05 s**. The percentage of Rayleigh time step is calculated automatically (~5%)

Set the simulation time and data write-out frequency

1. In the Simulation Time section, set the Total Time to **2s**
2. Set the Target Save Interval to **0.01s** to specify the write-out frequency

Step 2: Set the Solver Engine

- Choose either the CPU or GPU solver engine
- If CPU is chosen, set the grid size to 3 Rmin

Step 3: Run the Simulation

Simulation run time depends on the number of Cores or GPU hardware used.

1. Click the start Progress button  at the bottom of the simulation window
2. In Viewer Controls, click the **Refresh Viewer** button  or enable **Auto Update**  at any point to update the Viewer and see how the simulation is progressing

EDEM Analyst: Analyzing the Bonds

Configure the geometry

1. Switch to the Analyst
2. Expand Display, select Geometries and set the Opacity of the geometry to **0.7**
3. Hide the particle factory cylinder by selecting it then de-selecting “display particle factory”

Configure the particles

1. Select Particles and Color By, there are two options:

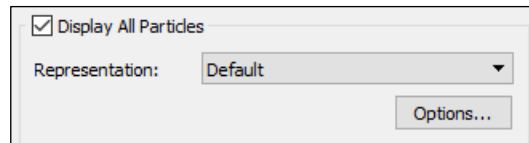
customBondsInitial – Indicates the number of bonds that an element has after the initial bond formation time.

customBondsDamage – Indicates the metric for the damage to the bonds for a given particle, calculated as:

$$1.0 - \frac{\text{current number of bonds for the particle}}{\text{customBondsInitial}}$$

A value of 0.0 indicates all initial bonds are still formed. A value of 1.0 indicates all initial bonds are now broken.

2. In the Particles section uncheck **Display All Particles** and click **Apply All**. The particles will disappear from the Viewer.



Configure the bonds

1. Select **Contact** from the Display Section of the Analyst Tree and choose the following settings:



2. You can color the contact by the following metrics:

customInitialOverlap – Indicates the initial value of the overlap of the elements in contact before the bond is formed.

BondStatus – Indicates the property that determines whether the bond exists or not. A value of 1.0 indicates a successful bond has been formed and is currently active. A value of -1.0 indicates that the bond has been broken. A value of 0.0 indicates a bond was never formed.

